

ASSIGNMENT IN MULTIVARIATE METHODS ST3101, Spring 2026

Due date 1: Saturday 21st March at 17.00

Due date 2: Tuesday 7th April at 17.00

Mode: Individual (collaboration is allowed but the written report is an individual effort)¹

Marks: Maximum 20 points and pass mark of 10 points

*If the assignment is not passed (a mark of F or Fx) after the first round of assessment (Due date 1) there is a possibility to revise the report. Revisions should be handed in by Due date 2. Note that it is **not** possible to revise the report if it is not passed after due date 2.*

Summary

The hand in assignment consists of performing **two** (2) multivariate analyses and writing a short report about your work. The report will be graded by your teacher and contribute to your final grade on the course.

Aims

You have to find a multivariate dataset, i.e. observations that consists of a reasonably large number of variables (say, greater than 3), motivate and use a combination of *two* of:

- Principal Components Analysis
- Factor Analysis (explorative or confirmatory)
- Cluster Analysis
- Classification using Discriminant Analysis
- Classification using Logistic Regression

Format

You shall use Rmarkdown and will hand in:

- A written report in .pdf *or* .html format
- The .Rmd file which has produced the report
- Any data files necessary for compiling the .Rmd file

Further, detailed guidelines for the format of the assignment and sources of data are provided in a separate .Rmd document.

¹Note that your submission may be run through text matching software to discover plagiarism. See course description for AI usage